Phenylephrine (systemic): Drug information - UpToDate

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Contributor Disclosures

For additional information see "Phenylephrine (systemic): Patient drug information" and "Phenylephrine (systemic): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Biorphen;
- FT Nasal Decongestant PE [OTC];
- Immphentiv;
- Medi-Phenyl [OTC];
- Non-Pseudo Sinus Decongestant [OTC];
- Sudafed PE Childrens [OTC];
- Sudafed PE Sinus Congestion [OTC];
- Sudanyl PE [OTC];
- Vazculep [DSC]

Pharmacologic Category

- Alpha-Adrenergic Agonist

Dosing: Adult

Hypotension or shock

Hypotension or shock: Note: Not recommended for septic shock *except* in the following circumstances: 1) when norepinephrine (preferred first-line agent) is associated with serious arrhythmias; 2) when cardiac output is known to be high and BP persistently low; or 3) used as salvage therapy when the combination of vasopressor/inotropic agents and low-dose vasopressin fail to achieve target mean arterial pressure (MAP) (Ref). In general, maintain goal MAP (eg, ~65 mm Hg); consider use if patient is in shock or has hypoperfusion during or after fluid resuscitation (Ref). Institutional protocols may vary with weight-based or non–weight-based dose regimens.

Septic shock and other vasodilatory shock states (alternative agent):

Weight-based dosing: **Continuous infusion: IV:** Initial dose: 0.5 to 2 mcg/kg/minute; titrate to desired MAP; usual dosage range: 0.25 to 5 mcg/kg/minute. Doses up to 9.1 mcg/kg/minute have been reported (Ref).

Non–weight-based dosing (based on ~80 kg patient): **Continuous infusion: IV:** Initial dose: 40 to 160 mcg/minute; titrate to desired MAP; usual dosage range: 20 to 400 mcg/minute. Doses up to

~730 mcg/minute have been reported (doses calculated and rounded for an 80 kg patient according to weight-based dosing using the referenced sources (Ref)).

Cardiogenic shock: **Note:** Depending on hemodynamic variables, may consider for vasoactive management of cardiogenic shock due to aortic stenosis, mitral stenosis, or hypertrophic cardiomyopathy with left ventricular outflow tract obstruction (Ref).

Continuous infusion: IV: 0.1 to 10 mcg/kg/minute; titrate to clinical end point (Ref).

Hypotension during anesthesia

Hypotension during anesthesia:

IV bolus: Initial: 40 to 100 mcg/dose; may repeat every 1 to 2 minutes as needed; titrate to clinical end point; maximum total dose: 200 mcg.

Continuous infusion: IV: Initial dose: 10 to 35 mcg/minute; titrate to clinical end point; maximum dose: 200 mcg/minute.

Nasal congestion

Nasal congestion: Oral: OTC labeling: 10 mg every 4 hours as needed for ≤7 days (maximum: 60 mg/24 hours).

Priapism, acute ischemic

Priapism, acute ischemic (off-label use): Note: The optimal dosing, frequency, and method of administration has not been established.

Intracavernous (off-label route): 100 to 500 mcg every 5 minutes or greater as needed for up to 1 hour (using a concentration of 100 to 500 mcg/mL); if used with aspiration/irrigation, then aspiration should be performed before administration of phenylephrine (Ref).

Dosing: Kidney Impairment: Adult

Mild to severe impairment: There are no dosage adjustments provided in the manufacturer's labeling.

End-stage kidney disease: There are no dosage adjustments provided in the manufacturer's labeling; dose-response data indicate increased responsiveness; lower initial doses may be required.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; dose-response data indicate decreased responsiveness; higher doses may be required.

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥30 kg/m^2):

Continuous infusion: IV: If institution uses weight-based dosing, use **ideal body weight** for initial weight-based dose calculations, then titrate to hemodynamic effect and clinical response (Ref). If institution uses nonweight-based dosing for vasoactive agents, continue with this approach. During therapy, clinicians should **not** change dosing weight from one weight metric to another (ie, ideal body weight to/from actual body weight or weight-based dosing to/from nonweight-based dosing) (Ref). Refer to adult dosing for indication specific doses.

Rationale for recommendations: There is a paucity of studies evaluating the influence of obesity on phenylephrine dosing or pharmacokinetics. Observational studies, including phenylephrine, epinephrine, and norepinephrine, suggest nonweight-based dosing strategies may result in lower overall cumulative dose requirements and increased drug exposure to second line agents in some patients and may not be advantageous in time to achieving hemodynamic stability (Ref). However, it is difficult to show outcome differences between weight-based and nonweight-based dosing because of dose titration to target BP, particularly in the context of retrospective studies. Furthermore, there is substantial variability in response in critically ill patients, irrespective of weight. Due to the short onset of action and small V_d, rapid titration to clinical effect after initial dosing is possible (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see ”Phenylephrine (systemic): Pediatric drug information”)

Dosage guidance:

Safety: Dosing presented in both mg (oral) and mcg (parenteral); use caution when ordering and dispensing.

Nasal congestion

Nasal congestion:

Note: FDA has determined oral phenylephrine to be ineffective for temporary relief of nasal decongestion; there is currently a proposed order to remove oral phenylephrine from the US market (Ref).

Children 4 to 5 years: Oral: 2.5 mg every 4 hours as needed; maximum daily dose: 15 mg in 24 hours.

Children 6 to 11 years: Oral: 5 mg every 4 hours as needed; maximum daily dose: 30 mg in 24 hours.

Children ≥12 years and Adolescents: Oral: 10 mg every 4 hours as needed; maximum daily dose: 60 mg in 24 hours.

Hypotension, low cardiac output

Hypotension, low cardiac output: Limited data available:

Infants, Children, and Adolescents:

IM, SUBQ: 100 mcg/kg/dose every 1 to 2 hours as needed; maximum dose: 5,000 mcg/dose (Ref).

IV bolus: 5 to 20 mcg/kg/dose every 10 to 15 minutes as needed (Ref); initial dose should not exceed 500 mcg/dose (Ref). **Note:** Start at lower dose; may repeat or increase dose as clinically indicated.

Continuous IV infusion: Usual initial dose: 0.1 to 0.5 mcg/kg/minute; titrate to desired response (Ref); in cases of shock, doses up to 2 mcg/kg/minute are recommended (Ref); higher doses up to 5 mcg/kg/minute may be required for management of infundibular spasms (tet spells) (Ref).

Dosing: Kidney Impairment: Pediatric

Mild to severe impairment: There are no dosage adjustments provided in the manufacturer’s labeling.

End-stage kidney disease: There are no dosage adjustments provided in the manufacturer’s labeling. Based on dose-response data (increased responsiveness) in adults, a lower initial dose is recommended.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer’s labeling. Based on dose-response data (decreased responsiveness) in adults, higher doses may be required.

Adverse Reactions

The following adverse drug reactions are derived from product labeling for IV use, unless otherwise specified.

Postmarketing:

Cardiovascular: Atrioventricular block, bradycardia, cardiac arrhythmia, cardiomyopathy (oral; takotsubo) (Zlotnick 2012), chest pain, hypertension, hypertensive crisis, ischemia, ischemic heart disease, low cardiac output, premature ventricular contractions, reflex bradycardia

Dermatologic: Diaphoresis, pallor, piloerection, pruritus

Gastrointestinal: Epigastric pain, ischemic colitis (oral) (El-Alali 2020), nausea, vomiting

Hypersensitivity: Hypersensitivity reaction (sulfite sensitivity; including anaphylaxis) (Pathan 2024))

Local: Localized blanching

Nervous system: Cerebrovascular accident (IV, oral; including cerebral hemorrhage) (Tark 2014), excitability, headache, nervousness, paresthesia, tremor

Neuromuscular & skeletal: Neck pain

Ophthalmic: Blurred vision

Respiratory: Dyspnea, pulmonary edema, rales

Contraindications

Immphentiv, premixed bag: Hypersensitivity to phenylephrine or any component of the formulation.

OTC labeling (Oral): When used for self-medication: Use with or within 14 days of monoamine oxidase inhibitor therapy.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Warnings/Precautions

Concerns related to adverse effects:

• Cardiovascular effects: IV use of phenylephrine may cause severe bradycardia (likely baroreflex mediated) and reduced cardiac output due to an increase in cardiac afterload especially in patients with preexisting cardiac dysfunction (Goertz 1993; Yamazaki 1982). May also precipitate angina in patients with severe coronary artery disease and increase pulmonary arterial pressure. Use with caution in patients with preexisting bradycardia, partial heart block, myocardial disease, or severe coronary artery disease. Avoid or use with extreme caution in patients with heart failure or cardiogenic shock; increased systemic vascular resistance may significantly reduce cardiac output. Avoid use in patients with hypertension (contraindicated in severe hypertension); monitor BP closely and adjust infusion rate. May also cause excessive peripheral and visceral vasoconstriction and ischemia to vital organs, particularly in patients with extensive peripheral vascular disease.

• Extravasation: IV administration: Vesicant; ensure proper needle or catheter placement prior to and during infusion. Avoid extravasation.

Disease-related concerns:

• Acidosis: Acidosis may reduce the efficacy of phenylephrine; correct acidosis during use of phenylephrine.

• Autonomic dysfunction: Patients with autonomic dysfunction (eg, spinal cord injury) may exhibit an exaggerated increase in BP response to phenylephrine.

Concurrent drug therapy issues:

• Monoamine oxidase inhibitors: Use with extreme caution in patients taking monoamine oxidase inhibitors; hypertension may result from concurrent use.

Dosage form specific issues:

• Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [”Inactive” 1997]; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer’s labeling.

• Oral: When used for self-medication (OTC), use with caution in patients with asthma, bowel obstruction/narrowing, hyperthyroidism, diabetes mellitus, cardiovascular disease, ischemic heart disease, hypertension, increased intraocular pressure, prostatic hyperplasia, or in older adults. Notify health care provider if symptoms do not improve within 7 days or are accompanied by fever. Discontinue and contact health care provider if nervousness, dizziness, or sleeplessness occur. FDA has determined oral phenylephrine to be ineffective for temporary relief of nasal decongestion; there is currently a proposed order to remove oral phenylephrine from the US market (FDA 2024).

• Sulfites: Some products contain sulfites, which may cause allergic reactions in susceptible individuals.

Other warnings/precautions:

• Appropriate use: When used IV in patients who are hypotensive, assure adequate circulatory volume to minimize need for vasoconstrictors.

Warnings: Additional Pediatric Considerations

Safety and efficacy for the use of cough and cold products in pediatric patients <4 years of age is limited; the AAP warns against the use of these products for respiratory illnesses in young children. Serious adverse effects including death have been reported. Many of these products contain multiple active ingredients, increasing the risk of accidental overdose when used with other products. The FDA does not recommend OTC uses for these products in pediatric patients <2 years of age and recommends to use with caution in pediatric patients ≥2 years of age. Health care providers are reminded to ask caregivers about the use of OTC cough and cold products in order to avoid exposure to multiple medications containing the same ingredient (AAP 2018; CDC 2007; FDA 2017; FDA 2018).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous, as hydrochloride:

Immphentiv: 0.5 mg/5 mL (5 mL); 1 mg/10 mL (10 mL) [contains edetate (edta) disodium]

Vazculep: 10 mg/mL (1 mL [DSC], 5 mL [DSC], 10 mL [DSC]) [contains sodium metabisulfite]

Generic: 10 mg/mL (1 mL, 5 mL, 10 mL)

Solution, Intravenous, as hydrochloride [preservative free]:

Biorphen: 0.5 mg/5 mL (5 mL)

Biorphen: 0.5 mg/5 mL (5 mL [DSC]) [sulfate free]

Generic: 0.5 mg/5 mL (5 mL); 10 mg/mL (1 mL, 5 mL, 10 mL); 20 mg/250 mL in NaCl 0.9% (250 mL)

Solution, Oral, as hydrochloride:

Sudafed PE Childrens: 2.5 mg/5 mL (118 mL) [alcohol free, sugar free; contains edetate (edta) disodium, fd&c red #40 (allura red ac dye), sodium benzoate; berry flavor]

Tablet, Oral, as hydrochloride:

FT Nasal Decongestant PE: 10 mg [pseudoephedrine free; contains fd&c red #40 (allura red ac dye)]

Medi-Phenyl: 5 mg [pseudoephedrine free; contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow)]

Non-Pseudo Sinus Decongestant: 10 mg [contains fd&c red #40(allura red ac)aluminum lake, fd&c yellow #6(sunset yellow)alumin lake]

Sudafed PE Sinus Congestion: 10 mg [contains fd&c red #40(allura red ac)aluminum lake, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Sudanyl PE: 5 mg [antihistamine free, caffeine free, salt free, sugar free; contains fd&c red #40 (allura red ac dye), fd&c yellow #6 (sunset yellow)]

Generic: 10 mg [DSC]

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Biorphen Intravenous)

0.5 mg/5 mL (per mL): $1.20

Solution (Immphentiv Intravenous)

0.5 mg/5 mL (per mL): $1.20

1 mg/10 mL (per mL): $1.20

Solution (Phenylephrine HCl (PF) Intravenous)

0.5 mg/5 mL (per mL): $1.20

Solution (Phenylephrine HCl Intravenous)

10 mg/mL (per mL): $1.01 - $4.80

Solution (Phenylephrine HCl-NaCl Intravenous)

20MG/250ML 0.9% (per mL): $0.31

Solution (Sudafed PE Childrens Oral)

2.5 mg/5 mL (per mL): $0.05

Tablets (Sudafed PE Sinus Congestion Oral)

10 mg (per each): $0.41

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Generic: 10 mg/mL (1 mL, 2 mL, 5 mL, 10 mL)

Solution Prefilled Syringe, Intravenous, as hydrochloride:

Generic: 0.5 mg/10 mL (10 mL)

Administration: Adult

IV:

Hypotension/shock: May be administered via continuous infusion (after dilution or when using ready-to-use or premixed formulations); IV infusions require an infusion pump. When administering as a continuous infusion, central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line is not available, may administer for a short duration (<72 hours) through a peripheral IV catheter placed in a large vein at a proximal site (eg, in or proximal to antecubital fossa). Frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation (Ref). Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution. Administration through midline catheters may also be an option (Ref).

Hypotension during anesthesia: Using 100 mcg/mL concentration, administer as an IV bolus over 20 to 30 seconds.

Intracavernous (off-label route): Using a 100 to 500 mcg/mL concentration, administer laterally (3 or 9 o'clock position) near the base of the penile shaft with small needles (eg, 27-gauge). May consider penile block with local anesthetic prior to administration (Ref).

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation.

Extravasation management: If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate phentolamine (or alternative antidote) (Ref).

Phentolamine: **SUBQ:** Dilute 5 to 10 mg in 10 mL 0.9% sodium chloride and administer into extravasation site as soon as possible after extravasation; if infiltrated catheter has not been removed, administer initial dose through the infiltrated catheter; may repeat in 60 minutes if patient remains symptomatic (Ref).

Alternative to phentolamine: Nitroglycerin topical 2% ointment: Apply a 1-inch strip to the site of ischemia to cover the affected area; may repeat every 8 hours as necessary (Ref).

Preparation for Administration: Adult

Note: Multiple formulations are available. The 100 mcg/mL (500 mcg per 5 mL and 1,000 mcg per 10 mL) Ready-to-use formulation does **not** require dilution prior to administration. The 10 mg/mL (10 mg/mL, 50 mg per 5 mL, and 100 mg per 10 mL) formulation **must be diluted** prior to administration. A 20 mg per 250 mL (80 mcg/mL) premixed bag that does not require dilution is also available.

Solution for injection:

IV infusion: May dilute 10 mg (1 mL of 10 mg/mL) in 500 mL D5W or NS to a concentration of 20 mcg/mL. May also dilute 50 mg in 500 mL NS, 40 mg in 500 mL NS, 100 mg in 500 mL NS, or 100 mg in 250 mL NS (Ref).

IV bolus injection: Dilute 10 mg (1 mL of 10 mg/mL) in 99 mL of D5W or NS to a concentration of 100 mcg/mL.

Intracavernous injection (off-label route): Dilute with NS to a concentration of 100 to 500 mcg/mL (Ref).

Usual Infusion Concentrations: Adult

IV infusion: 10 mg in 500 mL (concentration: 20 **mcg**/mL) of D5W or NS, 50 mg in 500 mL (concentration: 100 **mcg**/mL) of NS, 100 mg in 500 mL (concentration: 200 **mcg**/mL) of NS, **or** 100 mg in 250 mL (concentration: 400 **mcg**/mL) of NS.

Other institutions may use concentrations of 40 **mcg**/mL **or** 160 **mcg**/mL; however, stability information is not available for these concentrations.

Administration: Pediatric

Oral: OTC products: Administer without regard to food. Administer liquid formulation using an accurate measuring device; do not use household tablespoon.

Parenteral:

IV bolus: 10 mg/mL solution must be diluted prior to administration; a 100 mcg/mL (0.1 mg/mL) ready-to-use formulation requiring no further dilution is also available for IV bolus administration. Administer dose over 20 to 30 seconds (Ref).

Continuous IV infusion: 10 mg/mL solution must be diluted prior to administration. Administer as a continuous infusion via an infusion pump. Central line administration is preferred; extravasation may cause severe ischemic necrosis. If central line is not available, may consider administering for a short duration through a peripheral IV catheter placed in a large vein or via intraosseous access using a more dilute solution or with a second carrier fluid (Ref). Administration into an umbilical arterial catheter is **not** recommended. Frequent monitoring of the IV catheter site is recommended to rapidly identify extravasation. Refer to institutional policies and procedures; catheter placement/size and vasopressor concentration may vary depending on institution.

Vesicant; ensure proper needle or catheter placement prior to and during infusion; avoid extravasation. If extravasation occurs, stop infusion immediately; leave cannula/needle in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry, warm compresses. Initiate phentolamine (or alternative antidote) (Ref).

Preparation for Administration: Pediatric

Parenteral:

IV bolus: 10 mg/mL vial: Must be diluted prior to administration; may dilute with NS or D5W to a final concentration 100 mcg/mL (0.1 mg/mL) for bolus administration. **Note:** A 100 mcg/mL (0.1 mg/mL) ready-to-use formulation requiring no further dilution is also available.

Continuous IV infusion: 10 mg/mL vial: Must be diluted prior to administration; may dilute in NS or D5W to a final concentration of 20 mcg/mL per the manufacturer. ASHP recommends standard concentrations of 80 mcg/mL and 400 mcg/mL for pediatric patients (Ref).

Usual Infusion Concentrations: Pediatric

IV infusion: 20 **mcg**/mL, 80 **mcg**/mL, or 400 **mcg**/mL.

Vesicant/Extravasation Risk

Vesicant

Storage/Stability

Premixed bag (20 mg per 250 mL): Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Discard any unused portion.

Vials: Store intact vials at 20°C to 25°C (68°F to 77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F). Protect from light.

IV infusion: Concentrations of 0.1 and 0.2 mg/mL in NS are stable for at least 14 days at room temperature of 25°C (77°F) (Gupta 2004). The manufacturer's labeling for 10 mg/mL products state to store diluted solutions for ≤4 hours at room temperature or ≤24 hours refrigerated.

Stability in syringes: Concentration of 0.1 mg/mL in NS (polypropylene syringes) is stable for at least 30 days at -20°C (-4°F), 3°C to 5°C (37°F to 41°F), or 23°C to 25°C (73.4°F to 77°F) (Kiser 2007). Vazculep: Do not hold diluted solutions for longer than 4 hours at room temperature or 24 hours refrigerated.

Oral: Store at 15°C to 25°C (59°F to 77°F). Protect from light.

Use: Labeled Indications

Hypotension or shock: Treatment of severe hypotension or shock that persists during and after adequate fluid volume replacement.

Hypotension during anesthesia: Treatment of vasodilation and hypotension during anesthesia.

Nasal congestion [OTC]: For the temporary relief of nasal congestion due to the common cold, sinusitis, hay fever, or upper respiratory allergies.

Use: Off-Label: Adult

Priapism, acute ischemic

Medication Safety Issues

Sound-alike/look-alike issues:

Sudafed PE may be confused with Sudafed.

Vazculep may be confused with Bloxiverz (neostigmine) due to similar packaging.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (adrenergic agonist, IV; pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

International issues:

Biorphen: Brand name for phenylephrine systemic [US] but is also the brand name for orphenadrine [Great Britain]

Other safety concerns:

Neo (Neo-Synephrine, a well-known but DSC brand of phenylephrine) is an error-prone stemmed/coined drug name (mistaken as neostigmine)

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetaminophen: May increase serum concentrations of Phenylephrine (Systemic). *Risk C: Monitor*

Alpha1-Blockers: May decrease vasoconstricting effects of Phenylephrine (Systemic). *Risk C: Monitor*

Atomoxetine: May increase hypertensive effects of Sympathomimetics. Atomoxetine may increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Atropine (Systemic): May increase hypertensive effects of Alpha1-Agonists. *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Alpha1-Agonists and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Consider delaying skin testing until alpha1 agonists are no longer required and their effects have diminished, or use a histamine skin test as a positive control to assess a patient’s ability to mount a wheal and flare response. *Risk D: Consider Therapy Modification*

Bitter Orange: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Bornaprine: Sympathomimetics may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Bromocriptine: May increase hypertensive effects of Alpha1-Agonists. Management: Consider alternatives to this combination when possible. If combined, monitor for hypertension and tachycardia, and do not coadminister these agents for more than 10 days. *Risk D: Consider Therapy Modification*

Cannabinoid-Containing Products: May increase tachycardic effects of Sympathomimetics. *Risk C: Monitor*

Chloroprocaine (Systemic): May increase hypotensive effects of Phenylephrine (Systemic). *Risk C: Monitor*

CloZAPine: May decrease therapeutic effects of Phenylephrine (Systemic). *Risk C: Monitor*

Cocaine (Topical): May increase hypertensive effects of Sympathomimetics. Management: Consider alternatives to use of this combination when possible. Monitor closely for substantially increased blood pressure or heart rate and for any evidence of myocardial ischemia with concurrent use. *Risk D: Consider Therapy Modification*

Dihydralazine: Sympathomimetics may decrease therapeutic effects of Dihydralazine. *Risk C: Monitor*

Doxapram: Sympathomimetics may increase adverse/toxic effects of Doxapram. *Risk C: Monitor*

Doxofylline: Sympathomimetics may increase adverse/toxic effects of Doxofylline. *Risk C: Monitor*

Droxidopa: Sympathomimetics may increase hypertensive effects of Droxidopa. *Risk C: Monitor*

Ergot Derivatives (Vasoconstrictive CYP3A4 Substrates): May increase vasoconstricting effects of Alpha1-Agonists. *Risk X: Avoid*

Esketamine (Injection): May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for elevated heart rate, hypertension, and arrhythmias may be increased. *Risk C: Monitor*

FentaNYL: Decongestants may decrease serum concentrations of FentaNYL. *Risk C: Monitor*

Guanethidine: May increase hypertensive effects of Sympathomimetics. Guanethidine may increase arrhythmogenic effects of Sympathomimetics. *Risk C: Monitor*

Hyaluronidase: May increase vasoconstricting effects of Phenylephrine (Systemic). Management: Do not use hyaluronidase to enhance the dispersion or absorption of phenylephrine. Use of hyaluronidase for other purposes in patients receiving phenylephrine may be considered as clinically indicated. *Risk D: Consider Therapy Modification*

Iobenguane Radiopharmaceutical Products: Alpha1-Agonists may decrease therapeutic effects of Iobenguane Radiopharmaceutical Products. Management: Discontinue all drugs that may inhibit or interfere with catecholamine transport or uptake for at least 5 biological half-lives before iobenguane administration. Do not administer these drugs until at least 7 days after each iobenguane dose. *Risk X: Avoid*

Kratom: May increase adverse/toxic effects of Sympathomimetics. *Risk X: Avoid*

Landiolol: Sympathomimetics may decrease therapeutic effects of Landiolol. *Risk C: Monitor*

Linezolid: May increase hypertensive effects of Sympathomimetics. Management: Consider initial dose reductions of sympathomimetic agents, and closely monitor for enhanced blood pressure elevations, in patients receiving linezolid. *Risk D: Consider Therapy Modification*

Lisuride: May increase hypertensive effects of Alpha1-Agonists. Management: Consider alternatives when possible, as product labeling states the concomitant use of lisuride with alpha1-agonists is not recommended. If combined, monitor for increases in blood pressure or hypertensive crises. *Risk D: Consider Therapy Modification*

Melperone: May decrease therapeutic effects of Phenylephrine (Systemic). *Risk C: Monitor*

Metergoline: May increase adverse/toxic effects of Alpha1-Agonists. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase hypertensive effects of Alpha1-Agonists. While linezolid is expected to interact via this mechanism, management recommendations differ from other monoamine oxidase inhibitors. Refer to linezolid specific monographs for details. *Risk X: Avoid*

Pergolide: May increase hypertensive effects of Alpha1-Agonists. *Risk C: Monitor*

Propacetamol: May increase serum concentrations of Phenylephrine (Systemic). Management: Monitor patients closely for increased side effects of phenylephrine if propacetamol is used concomitantly. Patients with underlying blood pressure issues or arrhythmias may need closer monitoring and may warrant consideration of alternative therapies. *Risk C: Monitor*

Solriamfetol: Sympathomimetics may increase hypertensive effects of Solriamfetol. Sympathomimetics may increase tachycardic effects of Solriamfetol. *Risk C: Monitor*

Sympathomimetics: May increase adverse/toxic effects of Sympathomimetics. *Risk C: Monitor*

Tedizolid: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk for increased blood pressure and heart rate may be increased. *Risk C: Monitor*

Thyroid Products: May increase adverse/toxic effects of Sympathomimetics. Specifically, the risk of hypertension, tachycardia, and coronary insufficiency may be increased in patients with coronary artery disease. *Risk C: Monitor*

Tricyclic Antidepressants: May increase therapeutic effects of Alpha1-Agonists. Tricyclic Antidepressants may decrease therapeutic effects of Alpha1-Agonists. *Risk C: Monitor*

Pregnancy Considerations

Phenylephrine crosses the placenta (Ngan Kee 2009).

Outcome data are available following maternal use of phenylephrine as a decongestant (route of administration not always noted) (Aselton 1985; Heinonen 1977; Rothman 1979; Yau 2013) or for hypotension during cesarean delivery (Fitzgerald 2020; Heesen 2014; Singh 2020; Xu 2018).

Phenylephrine is available over the counter for the symptomatic relief of nasal congestion. Decongestants are not the preferred agents for the treatment of rhinitis during pregnancy. Oral phenylephrine should be avoided during the first trimester of pregnancy due to the potential for adverse pregnancy outcomes (AAAAI/ACAAI [Dykewicz 2020]).

Untreated hypotension associated with spinal anesthesia during cesarean delivery may lead to maternal nausea, vomiting, and fetal acidosis (van Dyk 2022). Phenylephrine injection is recommended for the prevention and/or treatment of maternal hypotension associated with spinal anesthesia in patients undergoing cesarean delivery (ASA 2016; Kinsella 2018; Macones 2019; van Dyk 2022).

Breastfeeding Considerations

It is not known if phenylephrine is present in breast milk.

According to the manufacturer, the decision to breastfeed should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother.

Dietary Considerations

Some products may contain phenylalanine and/or sodium.

Monitoring Parameters

BP (or mean arterial pressure), heart rate, cardiac output (as appropriate); intravascular volume status; creatinine and urine output; peripheral perfusion; monitor site of infusion for blanching/extravasation.

Consult individual institutional policies and procedures.

Mechanism of Action

Potent, direct-acting alpha-adrenergic agonist with virtually no beta-adrenergic activity; produces systemic arterial vasoconstriction. Such increases in systemic vascular resistance may result in dose-dependent increases in systolic and diastolic blood pressure and reductions in heart rate and cardiac output (most noticeable in patients with preexisting cardiac dysfunction).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Blood pressure increase/vasoconstriction: IM, SubQ: 10 to 15 minutes; IV: Immediate

Nasal decongestant: Oral: 15 to 30 minutes (Kollar 2007)

Duration:

Blood pressure increase/vasoconstriction: IM: 1 to 2 hours; IV: ~15 to 20 minutes; SubQ: 50 minutes

Nasal decongestant: Oral: ≤4 hours (Kollar 2007)

Absorption: Oral: Erratic and incomplete (Kanfer 1993)

Distribution: V_d: Initial: 26 to 61 L; V_{dss}: 184 to 543 L (mean: 340 L) (Hengstmann 1982)

Metabolism: Hepatic via oxidative deamination (Oral: 24%; IV: 50%); Undergoes sulfation (Oral [mostly within gut wall]: 46%; IV: 8%) and some glucuronidation; forms inactive metabolites (Kanfer 1993)

Bioavailability: Oral: ≤38% (Hengstmann 1982; Kanfer 1993)

Half-life elimination: Alpha phase: ~5 minutes; Terminal phase: 2 to 3 hours (Hengstmann 1982; Kanfer 1993)

Time to peak: Oral: 0.75 to 2 hours (Kanfer 1993)

Excretion: Urine (mostly as inactive metabolites)

Patient Counseling Points

(For additional information see ”Phenylephrine (systemic): Patient drug information”)

What is this drug used for?

All oral products:

• It is used to treat nose stuffiness.

Injection:

• It is used to treat low blood pressure.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

• Dizziness or headache

• Feeling nervous and excitable

• Trouble sleeping

• Restlessness

• Upset stomach or throwing up

Injection:

• Sweating a lot

• Stomach pain

• Neck pain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

• High blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Chest pain or pressure or a fast heartbeat

• Shakiness

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Injection:

• Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain

• A heartbeat that does not feel normal

• Blurred eyesight

• Shortness of breath, a big weight gain, or swelling in the arms or legs

• Slow heartbeat

• Numbness or tingling in the hands or feet

• Redness, burning, pain, swelling, blisters, skin sores, or leaking of fluid where the drug is going into your body

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AT) Austria: Neosynephrine | Phenylephrin aguettant;
- (AU) Australia: Amcal nasal decongestant pe | Chemists Own Sinus Relief PE | Codral relief | Decongestant PE | Demazin nasal decongestant | Maxiclear sinus relief | Neo synephrine | Nyal nasal decongestant | Nyal nasal decongestant pe | Nyal sinus relief pe | Pharmacy care nasal decongestant pe | Phenylephrine bnm | Sinutab pe sinus & pain relief | Sudafed pe | Sudafed pe nasal decongestant | Terry white chemists sinus decongestant pe | Vicks actioncold nasal relief;
- (BD) Bangladesh: Vasculex;
- (BE) Belgium: Fenylefrine Unimedic | Phenylephrine aguettant;
- (CH) Switzerland: Phenylephrin aguettant | Phenylephrin labatec;
- (CO) Colombia: Fenilefrina hcl;
- (CZ) Czech Republic: Phenylephrine | Phenylephrine aguettant;
- (DE) Germany: Biorphen | Phenylephrin aguettant | Phenylephrin sintetica;
- (EE) Estonia: Biorphen;
- (ES) Spain: Fenilefrina aguettant | Fenilefrina Altan;

- (FI) Finland: Fenylefrin abcur | Fenylefrin aguettant | Fenylefrin stragen | Fenylefrin unimedic;
- (FR) France: Phenylephrine aguettant | Phenylephrine medac | Phenylephrine renaudin;
- (GB) United Kingdom: Phenylephrine;
- (HU) Hungary: Biorphen;
- (ID) Indonesia: Pheneven;
- (IE) Ireland: Phenylephrine;
- (IT) Italy: Fenilefrina aguettant;
- (KE) Kenya: Phenylephrine aguettant;
- (KW) Kuwait: Phenylephrine aguettant;
- (LV) Latvia: Biorphen;
- (NL) Netherlands: Biorphen | Fenylefrin altan | Fenylefrine Aguettant;
- (NO) Norway: Fenylefrin | Fenylefrin aguettant | Fenylefrin stragen | Fenylefrin unimedic;
- (NZ) New Zealand: Ethics nasal decongestant | Sudafed pe;
- (PA) Panama: Fenilefrina aguettant | Fenilefrina sintetica;
- (PH) Philippines: Sinudrin;
- (PL) Poland: Biorphen;
- (PR) Puerto Rico: Biorphen | Immphentiv | Medi-phenyl | Phenylephrine HCL | Sudafed pe sinus congestion | Sudogest pe;
- (PT) Portugal: Fenilefrina Altan | Fenilefrina biojam;
- (QA) Qatar: Phenver | Vazculep;
- (RO) Romania: Biorphen;
- (SA) Saudi Arabia: Phenylephrine;
- (SE) Sweden: Fenylefrin abcur | Fenylefrin aguettant | Fenylefrin stragen | Fenylefrin unimedic;
- (SG) Singapore: Phenylalpha;
- (TH) Thailand: Phenylephrine;
- (TN) Tunisia: Phenylephrine;
- (UG) Uganda: Phenylephrine aguettant;
- (ZA) South Africa: Biorphen | Phenylalpha

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